Exercise sheet 4

Linear Optimization (311.160), WS 2026/27

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Exercise 4.1

Consider $\{x : Ax = b,\ x \geq 0\}$ *with*

$$A = \begin{pmatrix} 2 & 1 & 3 & -4 \\ 1 & 2 & 2 & -1 \end{pmatrix}, \qquad b = \begin{pmatrix} 6 \\ 6 \end{pmatrix}.$$

1. Show that $x = (3, 1, 1, 1)^T$ *is feasible, but not a basic solution. 2. Show that* $y = (-3, -2, 4, 1)^T$ *satisfies* $Ay = 0$*. 3. Compute a feasible* $\widetilde{x} = x - \alpha y$ *with at least one zero component.*

1. Feasibility and basis test

$$\begin{aligned}
(Ax)_1 &= 2 \cdot 3 + 1 \cdot 1 + 3 \cdot 1 - 4 \cdot 1 && \text{substitute } x \\
&= 6 + 1 + 3 - 4 && \text{multiply} \\
&= 7 + 3 - 4 && \text{add the first two terms} \\
&= 10 - 4 && \text{add} \\
&= 6 && \text{subtract} \\
(Ax)_2 &= 1 \cdot 3 + 2 \cdot 1 + 2 \cdot 1 - 1 \cdot 1 && \text{substitute } x \\
&= 3 + 2 + 2 - 1 && \text{multiply} \\
&= 5 + 2 - 1 && \text{add the first two terms} \\
&= 7 - 1 && \text{add} \\
&= 6 && \text{subtract}
\end{aligned}$$

Every component of x is positive, so x is feasible.

$$\begin{aligned}
\det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} &= 2 \cdot 2 - 1 \cdot 1 && \text{columns 1 and 2} \\
&= 4 - 1 && \text{multiply} \\
&= 3 && \text{subtract}
\end{aligned}$$

There are two rows and two independent columns, hence $\operatorname{rank}(A) = 2$. A basis has two columns. A basic solution has its other two components equal to zero. The given x has four positive components, so it cannot be a basic solution. □

x is feasible, but not basic.

2. A direction in the kernel

$$\begin{aligned}
(Ay)_1 &= 2 \cdot (-3) + 1 \cdot (-2) + 3 \cdot 4 - 4 \cdot 1 && \text{substitute } y \\
&= -6 - 2 + 12 - 4 && \text{multiply} \\
&= -8 + 12 - 4 && \text{subtract} \\
&= 4 - 4 && \text{add} \\
&= 0 && \text{subtract} \\
(Ay)_2 &= 1 \cdot (-3) + 2 \cdot (-2) + 2 \cdot 4 - 1 \cdot 1 && \text{substitute } y
\end{aligned}$$

$$
\begin{aligned}
&= -3 - 4 + 8 - 1 && \text{multiply}\\
&= -7 + 8 - 1 && \text{subtract}\\
&= 1 - 1 && \text{add}\\
&= 0 && \text{subtract}
\end{aligned}
$$

Thus the coefficients of y give the required nonzero linear dependence. $\square$

$$\boxed{Ay = (0,0)^T.}$$

3. First zero component

$$
\begin{aligned}
\widetilde{x} &= x - \alpha y && \text{given direction}\\
&= (3,1,1,1)^T - \alpha(-3,-2,4,1)^T && \text{substitute } x, y\\
&= (3 + 3\alpha, 1 + 2\alpha, 1 - 4\alpha, 1 - \alpha)^T && \text{distribute}
\end{aligned}
$$

For $\alpha \geq 0$ the first two entries remain positive. The last two require

$$
\begin{aligned}
1 - 4\alpha &\geq 0 && \text{third entry}\\
1 &\geq 4\alpha && \text{add } 4\alpha\\
\tfrac{1}{4} &\geq \alpha && \text{divide by } 4\\
1 - \alpha &\geq 0 && \text{fourth entry}\\
1 &\geq \alpha && \text{add } \alpha\\
\alpha &= \min\{\tfrac{1}{4}, 1\} && \text{first boundary in positive direction}\\
&= \tfrac{1}{4} && \text{compare bounds}
\end{aligned}
$$

$$
\begin{aligned}
\widetilde{x}_1 &= 3 + 3 \cdot \tfrac{1}{4} && \text{substitute } \alpha\\
&= 3 + \tfrac{3}{4} && \text{multiply}\\
&= \tfrac{15}{4} && \text{add}\\
\widetilde{x}_2 &= 1 + 2 \cdot \tfrac{1}{4} && \text{substitute } \alpha\\
&= 1 + \tfrac{1}{2} && \text{multiply}\\
&= \tfrac{3}{2} && \text{add}\\
\widetilde{x}_3 &= 1 - 4 \cdot \tfrac{1}{4} && \text{substitute } \alpha\\
&= 1 - 1 && \text{multiply}\\
&= 0 && \text{subtract}\\
\widetilde{x}_4 &= 1 - \tfrac{1}{4} && \text{substitute } \alpha\\
&= \tfrac{3}{4} && \text{subtract}
\end{aligned}
$$

Check the equations and nonnegativity:

$$
\begin{aligned}
(A\widetilde{x})_1 &= 2 \cdot \tfrac{15}{4} + \tfrac{3}{2} + 3 \cdot 0 - 4 \cdot \tfrac{3}{4} && \text{substitute}\\
&= \tfrac{15}{2} + \tfrac{3}{2} + 0 - 3 && \text{multiply}\\
&= 9 + 0 - 3 && \text{add fractions}\\
&= 9 - 3 && \text{remove zero}\\
&= 6 && \text{subtract}\\
(A\widetilde{x})_2 &= \tfrac{15}{4} + 2 \cdot \tfrac{3}{2} + 2 \cdot 0 - \tfrac{3}{4} && \text{substitute}\\
&= \tfrac{15}{4} + 3 + 0 - \tfrac{3}{4} && \text{multiply}
\end{aligned}
$$

$$
\begin{aligned}
&= \tfrac{15}{4} + 3 - \tfrac{3}{4} && \text{remove zero}\\
&= 3 + 3 && \text{subtract fractions}\\
&= 6 && \text{add}
\end{aligned}
$$

All four entries are nonnegative.

$$\boxed{\alpha = \tfrac{1}{4}, \qquad \widetilde{x} = (\tfrac{15}{4}, \tfrac{3}{2}, 0, \tfrac{3}{4})^T.}$$

Exercise 4.2

Continue Exercise 4.1. Let I index the nonzero entries of $\widetilde{x}$. Find a nonzero dependence $\widetilde{y}_I$ of the columns indexed by I. Set $\overline{x}_i = \widetilde{x}_i$ outside I and $\overline{x}_i = \widetilde{x}_i - \varepsilon\widetilde{y}_i$ on I, with one more zero entry. Are $\widetilde{x}$ and $\overline{x}$ basic feasible?

4. Second reduction and basis classification

The support is $I = \{1, 2, 4\}$. Write a dependence with coefficients (u, v, w):

$$
\begin{aligned}
2u + v - 4w &= 0 && \text{first row of } A_I(u,v,w)^T = 0\\
v &= 4w - 2u && \text{isolate } v\\
u + 2v - w &= 0 && \text{second row}\\
u + 2(4w - 2u) - w &= 0 && \text{substitute } v\\
u + 8w - 4u - w &= 0 && \text{distribute}\\
-3u + 7w &= 0 && \text{collect terms}\\
7w &= 3u && \text{add } 3u\\
u &= \tfrac{7}{3}w && \text{divide by 3, swap sides}
\end{aligned}
$$

Choose $w = 3$:

$$
\begin{aligned}
u &= \tfrac{7}{3} \cdot 3 && \text{substitute } w\\
&= 7 && \text{multiply}\\
v &= 4 \cdot 3 - 2 \cdot 7 && \text{substitute } w, u\\
&= 12 - 14 && \text{multiply}\\
&= -2 && \text{subtract}
\end{aligned}
$$

$$\boxed{\widetilde{y}_I = (7, -2, 3)^T, \qquad \widetilde{y} = (7, -2, 0, 3)^T.}$$

Check the direction:

$$
\begin{aligned}
(A\widetilde{y})_1 &= 2 \cdot 7 + 1 \cdot (-2) + 3 \cdot 0 - 4 \cdot 3 && \text{substitute}\\
&= 14 - 2 + 0 - 12 && \text{multiply}\\
&= 12 + 0 - 12 && \text{subtract}\\
&= 12 - 12 && \text{remove zero}\\
&= 0 && \text{subtract}\\
(A\widetilde{y})_2 &= 1 \cdot 7 + 2 \cdot (-2) + 2 \cdot 0 - 1 \cdot 3 && \text{substitute}\\
&= 7 - 4 + 0 - 3 && \text{multiply}\\
&= 3 + 0 - 3 && \text{subtract}\\
&= 3 - 3 && \text{remove zero}
\end{aligned}
$$

$= 0$ subtract

For a positive step only entries 1 and 4 decrease:

$$\begin{aligned}
\varepsilon &= \min\{(15/4)/7, (3/4)/3\} && \text{positive direction components}\\
&= \min\{15/28, 1/4\} && \text{divide fractions}\\
&= \tfrac{1}{4} && \text{compare bounds}
\end{aligned}$$

$$\begin{aligned}
\overline{x}_1 &= \tfrac{15}{4} - \tfrac{1}{4}\cdot 7 && \text{substitute } \varepsilon\\
&= \tfrac{15}{4} - \tfrac{7}{4} && \text{multiply}\\
&= \tfrac{8}{4} && \text{subtract numerators}\\
&= 2 && \text{reduce}\\
\overline{x}_2 &= \tfrac{3}{2} - \tfrac{1}{4}\cdot(-2) && \text{substitute } \varepsilon\\
&= \tfrac{3}{2} + \tfrac{1}{2} && \text{multiply and subtract a negative}\\
&= 2 && \text{add}\\
\overline{x}_3 &= 0 && \text{outside } I\text{, unchanged}\\
\overline{x}_4 &= \tfrac{3}{4} - \tfrac{1}{4}\cdot 3 && \text{substitute } \varepsilon\\
&= \tfrac{3}{4} - \tfrac{3}{4} && \text{multiply}\\
&= 0 && \text{subtract}
\end{aligned}$$

Check:

$$\begin{aligned}
(A\overline{x})_1 &= 2\cdot 2 + 1\cdot 2 + 3\cdot 0 - 4\cdot 0 && \text{substitute}\\
&= 4 + 2 + 0 - 0 && \text{multiply}\\
&= 4 + 2 && \text{remove zero terms}\\
&= 6 && \text{add}\\
(A\overline{x})_2 &= 1\cdot 2 + 2\cdot 2 + 2\cdot 0 - 1\cdot 0 && \text{substitute}\\
&= 2 + 4 + 0 - 0 && \text{multiply}\\
&= 2 + 4 && \text{remove zero terms}\\
&= 6 && \text{add}
\end{aligned}$$

The point $\widetilde{x}$ has three positive entries and cannot be basic at rank two. For $\overline{x}$, choose $B = (1, 2)$ and set $x_3 = x_4 = 0$. The determinant of A_B is 3, as computed in 4.1, so this is a basis. Both remaining entries are positive. □

$\varepsilon = \frac{1}{4}, \qquad \overline{x} = (2, 2, 0, 0)^T,$
$\widetilde{x}$ is not basic feasible, $\qquad \overline{x}$ is basic feasible.

Exercise 4.3 (Programming)

For Example 2.2, maximize $-x_1 + 2x_2 + 6x_3$ subject to

$$2x_1 + 3x_2 + x_4 = 10, \qquad x_1 + 2x_3 + x_5 = 8, \qquad x_1 + 2x_2 + x_3 + x_6 = 6, \qquad x_1, \ldots, x_6 \geq 0.$$

Determine c, A, b and a basis B. Complete `initial_simplex_dictionary` *from the supplied skeleton and test it on these data. Return the dimensions, B, N, $\overline{b}$, $\overline{A}$, $\overline{z}$, reduced costs, basic solution and feasibility. Submit the Python and PDF files as* `LASTNAME_sheet_NUMBER.py` *and* `LASTNAME_sheet_NUMBER.pdf`.

Data and initial basis

$$c = \begin{pmatrix} -1 \\ 2 \\ 6 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 3 & 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 8 \\ 6 \end{pmatrix}.$$

Choose $B = (4, 5, 6)$ and $N = (1, 2, 3)$. The corresponding submatrices and costs are

$$A_B = I_3, \quad A_N = \begin{pmatrix} 2 & 3 & 0 \\ 1 & 0 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \quad c_B = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad c_N = \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix}.$$

The columns of I_3 are independent, so B is a basis.

Dictionary quantities

$\overline{b} = A_B^{-1} b$	definition
$= I_3 b$	inverse of the identity
$= (10, 8, 6)^T$	identity multiplication
$\overline{A} = A_B^{-1} A_N$	definition
$= I_3 A_N$	inverse of the identity
$= A_N$	identity multiplication
$\overline{z} = c_B^T \overline{b}$	definition
$= 0 \cdot 10 + 0 \cdot 8 + 0 \cdot 6$	substitute
$= 0 + 0 + 0$	multiply
$= 0$	add
$\overline{c}_N^T = c_N^T - c_B^T \overline{A}$	definition
$= (-1, 2, 6) - (0, 0, 0)\overline{A}$	substitute costs
$= (-1, 2, 6) - (0, 0, 0)$	zero row times matrix
$= (-1, 2, 6)$	subtract zero

$x_B = \overline{b} - \overline{A} x_N$	dictionary convention
$x_4 = 10 - 2x_1 - 3x_2$	first row
$x_5 = 8 - x_1 - 2x_3$	second row
$x_6 = 6 - x_1 - 2x_2 - x_3$	third row
$z = \overline{z} + \overline{c}_N^T x_N$	objective dictionary
$= 0 + (-1, 2, 6)(x_1, x_2, x_3)^T$	substitute
$= -x_1 + 2x_2 + 6x_3$	row times column

For the basic solution set $x_N = (0, 0, 0)^T$:

$x_4 = 10 - 2 \cdot 0 - 3 \cdot 0$	substitute
$= 10 - 0 - 0$	multiply
$= 10$	subtract zeros

$$
\begin{aligned}
x_5 &= 8 - 0 - 2 \cdot 0 && \text{substitute} \\
&= 8 - 0 - 0 && \text{multiply} \\
&= 8 && \text{subtract zeros} \\
x_6 &= 6 - 0 - 2 \cdot 0 - 0 && \text{substitute} \\
&= 6 - 0 - 0 - 0 && \text{multiply} \\
&= 6 && \text{subtract zeros}
\end{aligned}
$$

All entries of x_B are nonnegative. Check in the original system:

$$
\begin{aligned}
(Ax)_1 &= 2 \cdot 0 + 3 \cdot 0 + 10 && \text{substitute} \\
&= 0 + 0 + 10 && \text{multiply} \\
&= 10 && \text{add} \\
(Ax)_2 &= 0 + 2 \cdot 0 + 8 && \text{substitute} \\
&= 0 + 0 + 8 && \text{multiply} \\
&= 8 && \text{add} \\
(Ax)_3 &= 0 + 2 \cdot 0 + 0 + 6 && \text{substitute} \\
&= 0 + 0 + 0 + 6 && \text{multiply} \\
&= 6 && \text{add} \\
c^T x &= -0 + 2 \cdot 0 + 6 \cdot 0 && \text{substitute} \\
&= 0 + 0 + 0 && \text{multiply} \\
&= 0 && \text{add}
\end{aligned}
$$

$$
\boxed{(m, n) = (3, 6), \quad x = (0, 0, 0, 10, 8, 6)^T, \quad z = 0, \quad \texttt{is_feasible=True.}}
$$

Python implementation and test

The completed function is in `Bleier_sheet_4.py`. It retains the skeleton's keys and adds `x_N` and the full vector `x`. It solves the systems $A_B \bar{b} = b$ and $A_B \overline{A} = A_N$ with `numpy.linalg.solve`; this computes the stated formulas without forming an inverse explicitly. Indices in the function are zero-based.

Run `python Bleier_sheet_4.py` for Example 2.2 and `python -m unittest -v test_sheet4.py` for the tests. The example returns the quantities calculated above. The tests also reconstruct $Ax = b$ and the objective for a non-identity basis in a different column order, and check an infeasible basis, a degenerate feasible basis, invalid inputs and a singular basis.

$$
\boxed{\bar{b} = (10, 8, 6)^T, \quad \overline{A} = A_N, \quad \bar{z} = 0, \quad \bar{c}_N = (-1, 2, 6)^T.}
$$